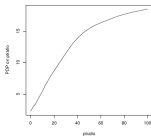
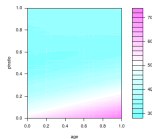
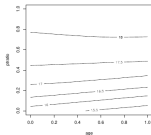
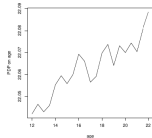
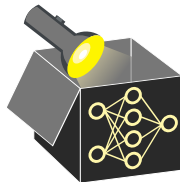


Interpretable Machine Learning

Functional Decompositions Further Methods

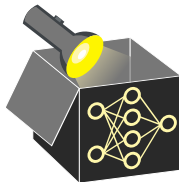


Learning goals

- Limitations of classical fANOVA
- Alternatives: Generalized fANOVA and ALE
- Advantages and relevance of functional decompositions

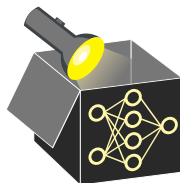
LIMITATIONS OF CLASSICAL FANOVA

- Standard fANOVA builds on PD-functions
- *Remember:* Problems of PDPs for **correlated / dependent features**
- Here: Dependent features $\implies$ Standard fANOVA does NOT fulfill vanishing conditions



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Example

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$$\hat{f}(x_1, x_2, x_3) = -2x_1 - 2\sin(x_3) + |x_1|x_2 + 0.5x_2x_3 + 1.$$

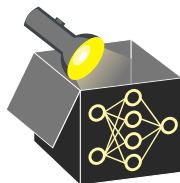
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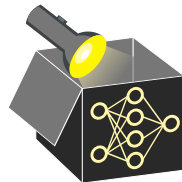
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$\rightarrow$ Extreme example, but again: Problem of definition

ALTERNATIVE: GENERALIZED FUNCTIONAL ANOVA

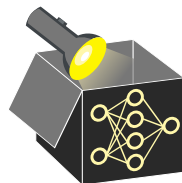
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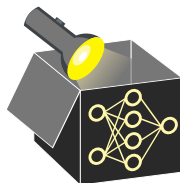
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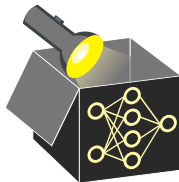
↪ Only components are orthogonal where $g_V(\mathbf{x}_V)$ is "lower in hierarchy" than $g_S(\mathbf{x}_S)$

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- **Advantage:** Also provides a variance decomposition



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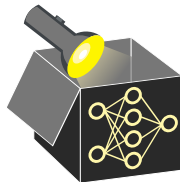
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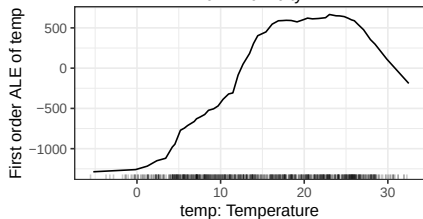
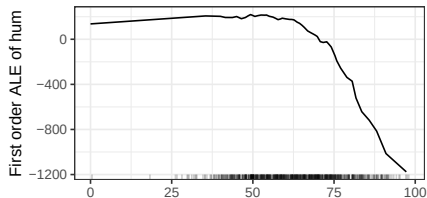
$\Rightarrow$ Generalized fANOVA provides functional decomp. for arbitrary settings

- **Advantage:** Also provides a variance decomposition
- **Problems:**
 - Difficult to estimate, involves manual choice of a "weight function"
 - Computationally very costly

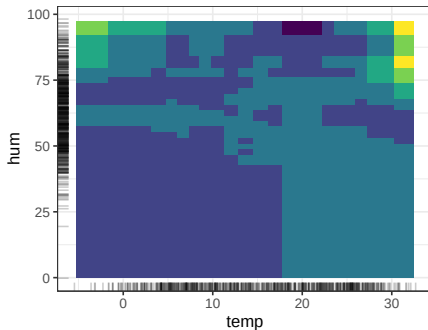
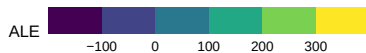
REVISITING ALE PLOTS



$$\hat{f}_{S,ALE}(x) = \sum_{k=1}^{k_S(x)} \frac{1}{n_S(k)} \sum_{i: x_S^{(i)} \in [z_{k-1,S}, z_{k,S}]} \left[\hat{f}(z_{k,S}, \mathbf{x}_{-S}^{(i)}) - \hat{f}(z_{k-1,S}, \mathbf{x}_{-S}^{(i)}) \right]$$



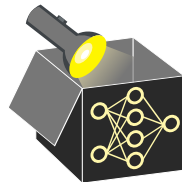
Second order ALE



ALE DECOMPOSITION

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→ Gives full functional decomposition of ALE plots



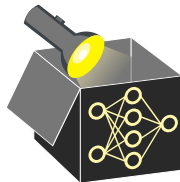
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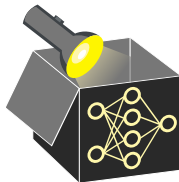
- **Advantages:** Handle dependencies well + computationally fast
- Constraints / orthogonality properties more complicated

⇒ ALE decomp. theoretically more involved, but good alternative in practice



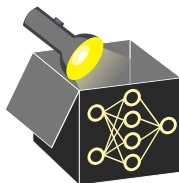
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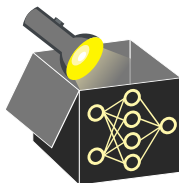


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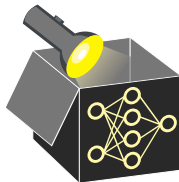
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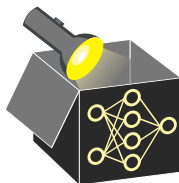


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Overall: Very important concept and theoretical background, explains idea behind many other methods